

Archaeologist

Genealogy · CLEAR vs TENTATIVE · intellectual move

Genealogy one-liner

Lab teleop (ALOHA) || DP chunks → UMI handheld wild teaching (human-in-wild) || DROID robot-in-wild → VLA/FM chunk policies ($\pi_0/\pi_{0.5}$) that still need UMI-class wrist-aligned demos.

Lineage diagram

Priors → UMI moment → descendants



Intellectual move

Flip “who goes into the wild” from robots to humans — while keeping robot-ready action (relative SE(3) + latency) so DP/VLA heads can deploy.

Priors UMI absorbs — CLEAR

What it stands on

Diffusion Policy

2303.04137

Generative action chunks + receding control — same first authors Chi/Xu/Song + TRI

ORB-SLAM3 + Urban GoPro

VIO stack

IMU-aware monocular SLAM for metric EE under fast motion

Handheld quasi-static grippers

GM / VIME / Dobb-E

UMI's foil — prior handheld limited to grasp/pick-place action poverty

Soft / series-elastic fingers

SEED / Alspach TRI

Continuous width + compliance via 95A TPU

ACT (named)

2304.13705

Potential drop-in chunk policy — CLEAR stated; equal empirical TENTATIVE

Contemporaneous contrast

ALOHA lab || DROID robot-wild || UMI human-wild

ALOHA/ACT

Lab puppet teleop · joint-space · embodiment-specific · fine bimanual

DROID

Mar 2024 neighbor · Franka Quest teleop in buildings · opposite “who enters wild”

Mobile ALOHA / GELLO

Still robot-present collection — not handheld GoPro pipe

UMI

Human-in-wild · relative EE · latency-matched DP · twin wrist cam on any arm

Descendants / citations

CLEAR vs TENTATIVE discipline

CLEAR

$\pi_{0.5}$ (2504.16054) bibliography cites UMI

CLEAR

GR00T-N1 (2503.14734) cites UMI

CLEAR

umi-on-legs · umi-arx deploy forks

CLEAR

UMI builds on Diffusion Policy (same authors)

TENTATIVE

ACT equal empirical drop-in (stated, not ablated)

TENTATIVE

π_0 /OpenVLA/Octo treat UMI as primary demo prior

TENTATIVE

BEHAVIOR'26 official baseline names UMI

UMI moment — numbers that stuck

What later papers inherit

Rel 100%

vs abs 25%

Lat 87.5%

vs 57.5%

Wild 71.7%

vs narrow 0%

Ta=6

@10/20 Hz

UR5→FR2

100→90%

73 + 298

portable BoM

What later papers learned

+ / - / systems

+ Relative EE + wrist alignment is the portable action alphabet for cross-robot deploy

+ Latency matching is not an implementation detail — it is a dynamic-task feature

+ Diverse handheld data beats pretrained ViT on narrow scenes (OOD 0%→71.7%)

— SLAM texture dependence remains a systems tax; kin-filter is post-hoc embodiment

— Parallel-jaw + 780g limits the “anyone can collect” story for marathon/dexterity

sys Open MIT stack + BoM made the interface citable and forkable (legs/ARX)

Fig. 2 — observation alignment as genealogy hinge

Twin wrist geometry is why relative policies transfer

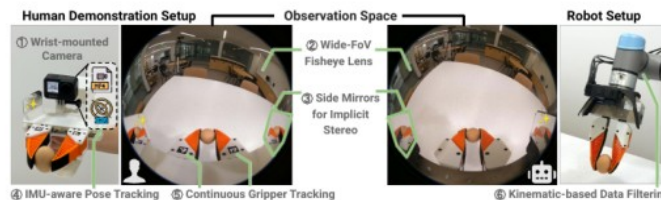


Fig. 2: UMI Demonstration Interface Design. Left: Hand-held grippers for data collection, with a GoPro as the only sensor and recording device. Middle: Image from the GoPro's 155° Fisheye view. Note the physical side mirrors highlighted in green which provide implicit stereo information. Right: UMI-compatible robot gripper and camera setup make observation similar to hand-held gripper view.

precise global action due to scale ambiguity, motion blur, or insufficient texture, which significantly restrict the precision of tasks for which the system can be employed.

- **Latency discrepancies:** During hand-held data collection, observation and action recording occur without latency. However, during inference, various latency sources, including sensor, inference, and execution latencies, arise within the system. Policies unaware of these latency discrepancies will encounter out-of-distribution input and in turn, generate out-of-sync actions. This issue is especially salient for fast and dynamic actions.

- **Insufficient policy representation:** Prior works often use simple policy representations (e.g., MLPs) with action regression loss, limiting their capacity to capture complex multimodal action distributions inherent in human data. Consequently, even with precisely recovered demonstrated actions and all discrepancies removed, the resulting policy could still struggle to fit the data accurately. This further hampers large-scale, distributed human data collection, as more demonstrators increase action multimodality.

In this paper, we address these issues with careful design of the demonstration and policy interface:

- First, we aim to identify the right **physical interface** for human demonstration that is intuitive and meanwhile able to capture all the information necessary for policy learning. Specifically, we use a Fisheye lens to increase the field of view and visual context, and add side mirrors on the gripper to provide implicit stereo observation. When combined with the GoPro's built-in IMU sensor, we can enable robust tracking under fast motion.
- Second, we explore the right **policy interface** (i.e., observation and action representations) that could make the policy hardware-agnostic and thereby enable effective skill transfer. Concretely, we employ inference-time latency matching to handle different sensor observation and execution latency, use relative trajectory as action representation to remove the need for precise global action, and finally, apply Diffusion Policy [9] to model

multimodal action distributions.

The final system, **Universal Manipulation Interface (UMI)**, provides a practical and accessible framework to unlock new robot manipulation skills, allowing us to demonstrate any actions in any environment while maintaining high transferability from human demonstration to robot policy.

With just a wrist-mounted camera on the hand-held gripper (Fig. 2), we show that UMI is capable of achieving a wide range of manipulation tasks that involve dynamic, bimanual, precise and long-horizon actions by only changing the training data for each task (Fig. 1). Furthermore, when trained with diverse human demonstrations, the final policy exhibits zero-shot generalization to novel environments and objects, achieving a remarkable 70% success rate in out-of-distribution tests, a level of generalizability seldomly observed in other behavior cloning frameworks. We open-source the hardware and software system at <https://umi-gripper.github.io>.

II. RELATED WORKS

A key enabler for any data-driven robotics system is the data itself. Here, we review a few typical data collection workflows in the context of robotic manipulation.

A. Teleoperated Robot Data

Imitation learning learns policies from expert demonstrations. Behavior cloning (BC), utilizing teleoperated robot demonstrations, stands out for its direct transferability. However, teleoperating real robots for data collection poses significant challenges. Previous approaches utilized interfaces such as 3D spacemouse [9, 54], VR or AR controllers [15, 3, 13, 19, 31, 51, 12], smartphones [44, 43, 22], and haptic devices [38, 47, 43, 26, 4] for teleoperation. These methods are either very expensive or hard to use due to high latency and lack of user intuitiveness. While recent advancements in leader-follower (i.e. puppeting) devices such as ALOHA [53, 15] and GELLO [46] offer promise with intuitive and low-cost interfaces, their reliance on real robots during data collection limits the type and number of environments the system can gain access to for “in-the-wild” data acquisition. Exoskeletons [14, 20] remove the dependence on real robots during data

Hardware interface is intellectual contribution, not just STL files

Into VLA / FM era

Why UMI still matters after π_0

π_0 / $\pi_{0.5}$ / OpenPI inherit chunked relative-ish control habits; they still need wrist-aligned demos.

UMI supplies the collection interface when robot-in-wild (DROID) is too expensive per scene.

GR00T cites UMI — humanoid stacks also need portable demo pipes.

Conceptual line: any wrist-cam + relative-EE + chunk policy stack (OpenPI recipes, BEHAVIOR finetunes).

Mark BEHAVIOR official baseline naming as TENTATIVE until organizer checklist confirms.

Exam bite — Archaeologist

UMI = in-the-wild WITHOUT in-the-wild robots. Genealogy: ALOHA||DP → UMI human-wild || DROID
robot-wild → $\pi_0/\pi_{0.5}$ that still need UMI-class wrist demos. CLEAR: DP prior; $\pi_{0.5}/GR00T$
cite. TENTATIVE: ACT equal; BEHAVIOR official name.

Numbers tattoo — pin these

Every deck carries the same spine numbers

100% / 90%

cup UR5 / Franka

87.5%

toss w/ latency

57.5%

toss no latency

71.7%

wild 43/60 OOD

0%

narrow → wild OOD

100% vs 25%

rel vs abs action

55%

no fisheye (69°)

70% / 30%

cloth ± inter-grip

70% / 0%

dish ViT / ResNet

1400 / 30

wild demos / locs

Ta=6

@10Hz (20Hz toss)

73 + 298

BOM · 780g

CLEAR vs TENTATIVE card

Do not blur these in exam answers

CLEAR

- UMI $\leftarrow$ Diffusion Policy
- vs DROID / ALOHA contrast
- $\pi_{0.5}$ / GR00T cite UMI
- Relative + latency + fisheye
- Open MIT stack / BoM

TENTATIVE / FORBIDDEN

- ACT equal empirical swap
- π_0 /OpenVLA primary-prior
- BEHAVIOR'26 official baseline
- Invented GPU wall-clock hours
- External third-person cams in UMI

Archaeologist — close

Takeaways + exam bite

1. Genealogy: ALOHA||DP $\rightarrow$ UMI human-wild || DROID robot-wild $\rightarrow \pi_0/\pi_{0.5}$.
2. CLEAR: DP prior; $\pi_{0.5}$ /GR00T cite; vs DROID/ALOHA contrast.
3. TENTATIVE: ACT equal; BEHAVIOR official baseline name.
4. Intellectual move: flip who enters the wild — keep robot-ready actions.

EXAM BITE

UMI=wild without robots; CLEAR DP+ $\pi_{0.5}$ /GR00T; TENTATIVE ACT equal & BEHAVIOR name.